

MTM1532 INT. TO FINANCIAL MATHEMATICS

Review for the Lecture 6 and the Lecture 7

(Questions are taken from Problem Sets)

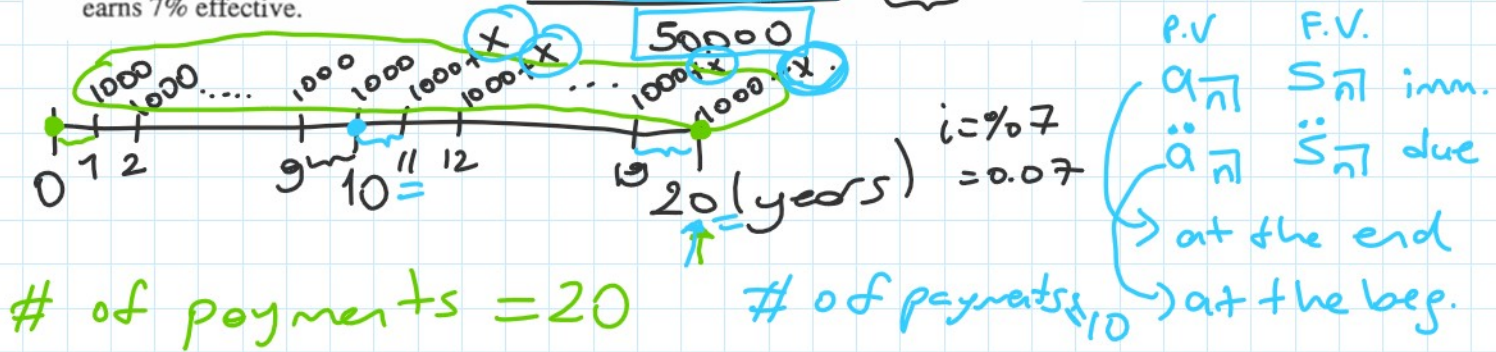
Text Book:

KELLISON, S. G., 1991, The Theory of Interest, Irwin Inc., USA.

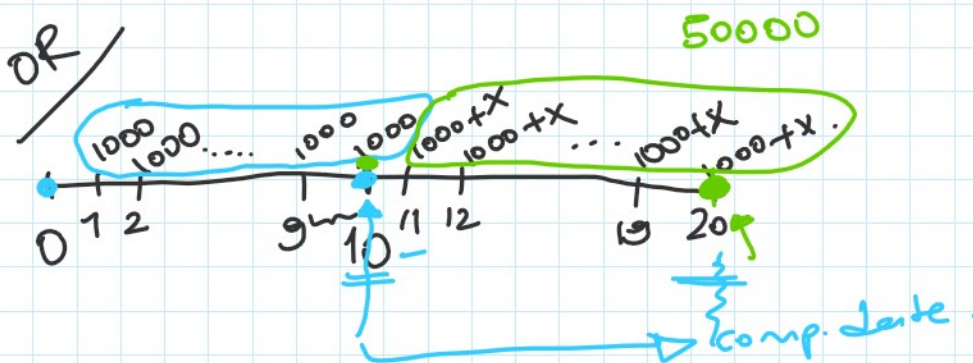
MTM1532 Review (Lec 6, Lec 7)

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1. A family wishes to accumulate \$50,000 in a college education fund at the end of 20 years. If they deposit \$1000 in the fund at the end of each of the first 10 years and \$1000 + X in the fund at the end of each of the second 10 years, find X if the fund earns 7% effective.



$$1000 \cdot S_{20|0.07} + X \cdot S_{10|0.07} = 50,000$$



$$1000 \cdot S_{10|0.07} \cdot (1 + 0.07)^{10} + (1000 + X) \cdot S_{10|0.07} = 50,000$$

The equation of value using a comparison date at time $t = 20$ is

$$50,000 = 1000s_{20|} + Xs_{10|} \text{ at } 7\%.$$

Thus,

$$X = \frac{50,000 - 1000s_{20|}}{s_{10|}} = \frac{50,000 - 40,995.49}{13.81645} = \$651.72.$$

Please verify. $S_n = \frac{(1+i)^n - 1}{i}$

2. The cash price of an automobile is \$10,000. The buyer is willing to finance the purchase at 18% convertible monthly and to make payments of \$250 at the end of each month for four years. Find the down payment that will be necessary.

10000

250 250 ... 250 250 ann-imm.

0 1 2 ... 47 48(month)
4 year.

$i^{(12)} = 0.18$

$4 \text{ year} \times 12 = 48$

$250 \cdot a_{\overline{48}|j} = P.V.$ where $j = \frac{i^{(12)}}{12} = \frac{0.18}{12} = \dots$

Down payment = $10000 - P.V$

The down payment (D) plus the amount of the loan (L) must equal the total price paid for the automobile. The monthly rate of interest is $j = .18/12 = .015$ and the amount of the loan (L) is the present value of the payments, i.e.

$$L = 250a_{\overline{48}|.015} = 250(34.04255) = 8510.64 = PV$$

Thus, the down payment needed will be

$$D = 10,000 - 8510.64 = \$1489.36 //$$

3. A sports car enthusiast needs to finance \$25,000 of the total purchase price of a new car. A loan is selected having 48 monthly level payments with a lender charging 6% convertible monthly. However, the lender informs the buyer that their policy is not to exceed a \$500 monthly payment on any car loan. The buyer decides to accept the loan offer with the \$500 payment and then decides to take out a second 12-month loan with a different lender at 7.5% convertible monthly to make up the shortfall not covered by the first loan. Find the amount of the monthly payment on the second

25000

X X X ... X

500 500 500 ... 500 500

0 1 2 3 ... 12 ... 47 48(month)

$i^{(12)} = 0.075$

$j_2 = \frac{0.075}{12}$

$i^{(12)} = 0.06$

$j_1 = \frac{0.06}{12} = 0.005$

$25000 = 500 \cdot a_{\overline{48}|j_1} + X \cdot a_{\overline{12}|j_2}$

$$a_{\overline{48}|j_1} = \dots$$

$$a_{\overline{12}|j_2} = \dots$$

} Solve for X.

The monthly interest rate on the first loan (L_1) is $j_1 = .06/12 = .005$ and

$$L_1 = 500a_{\overline{48}|.005} = (500)(42.58032) = 21,290.16.$$

The monthly interest rate on the second loan (L_2) is $j_2 = .075/12 = .00625$ and

$$L_2 = 25,000 - L_1 = 25,000 - 21,290.16 = 3709.84.$$

The payment on the second loan (R) can be determined from

$$3709.84 = Ra_{\overline{12}|.00625}$$

$$R = X.$$

giving

$$R = \frac{3709.84}{11.52639} = \$321.86.$$

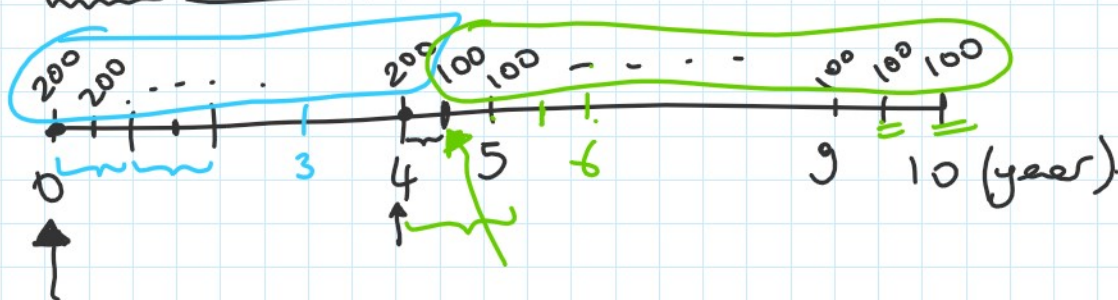
8. Find the present value of payments of \$200 every six months starting immediately and continuing through four years from the present, and \$100 every six months thereafter through ten years from the present, if $i^{(2)} = .06$.

ann-due.

$m \rightarrow 1 \text{ year.}$

$$i^{(2)} = 0.06$$

$$\frac{i^{(2)}}{2} = j = 0.03$$



of payments = 9

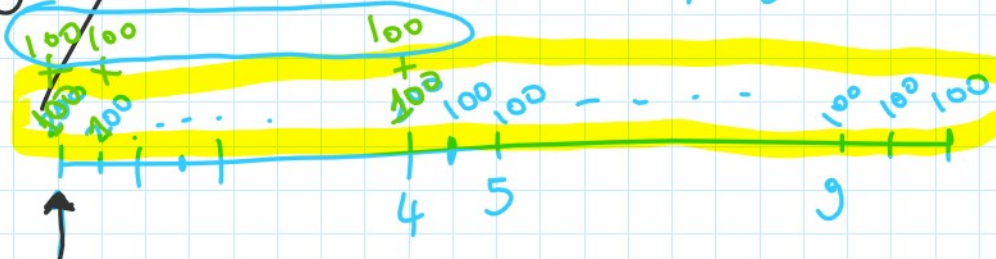
of payments = 12

$$200 \cdot \ddot{a}_{\overline{9}|j} + 100 \cdot a_{\overline{12}|j} \cdot (1+j)^{-2 \times 4}$$

$$200 \cdot \ddot{a}_{\overline{9}|j} + 100 \cdot \ddot{a}_{\overline{12}|j} \cdot (1+j)^{-2 \times (4.5)}$$

OR

of payments = 9.



of payments = 21

↑

4 ✓

✓

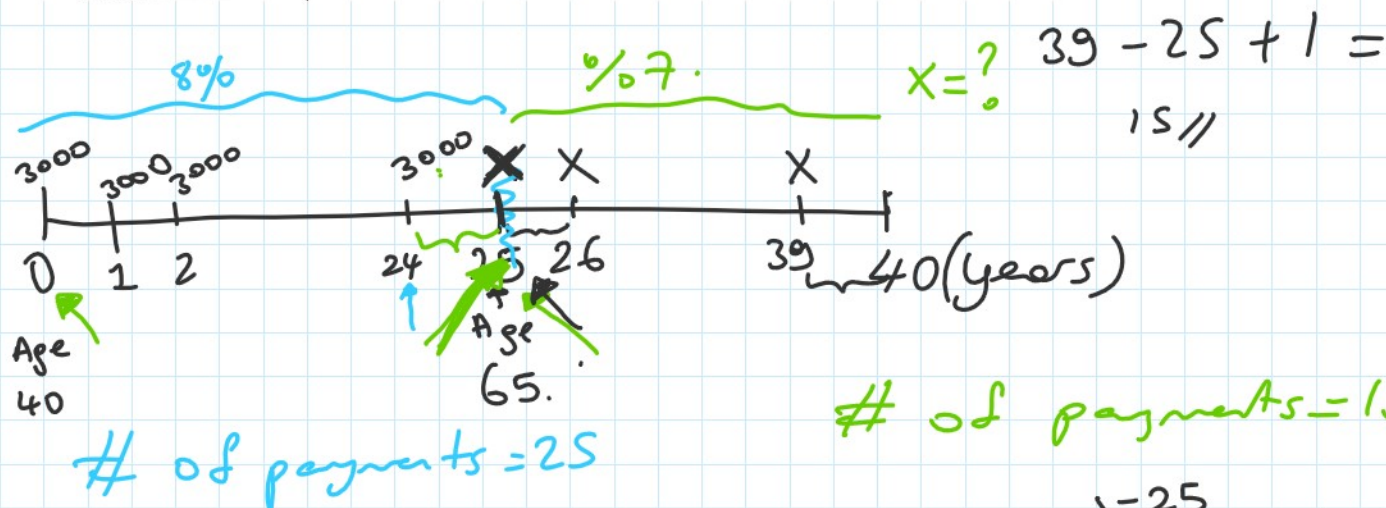
1 ✓
= 21

$$100 \cdot \ddot{a}_{\overline{9}|j} + 100 \cdot \ddot{a}_{\overline{21}|j}$$

The semiannual interest rate is $j = .06/2 = .03$. The present value of the payments is

$$100(\ddot{a}_{\overline{9}|.03} + \ddot{a}_{\overline{21}|.03}) = 100(15.87747 + 8.01969) = \$2389.72.$$

9. A worker aged 40 wishes to accumulate a fund for retirement by depositing \$3000 at the beginning of each year for 25 years. Starting at age 65 the worker plans to make 15 annual withdrawals at the beginning of each year. Assuming that all payments are certain to be made, find the amount of each withdrawal starting at age 65 to the nearest dollar, if the effective rate of interest is 8% during the first 25 years but only 7% thereafter.



$$3000 \cdot \ddot{s}_{\overline{25}|0.08} = X \cdot \ddot{a}_{\overline{15}|0.07} \cdot (1+0.08)^{-25}$$

comparision date is 0.

or if the comp. date is $t=25$ ~~HW!~~ verify.

$$3000 \cdot \ddot{s}_{\overline{25}|0.08} = X \cdot \ddot{a}_{\overline{15}|0.07}$$

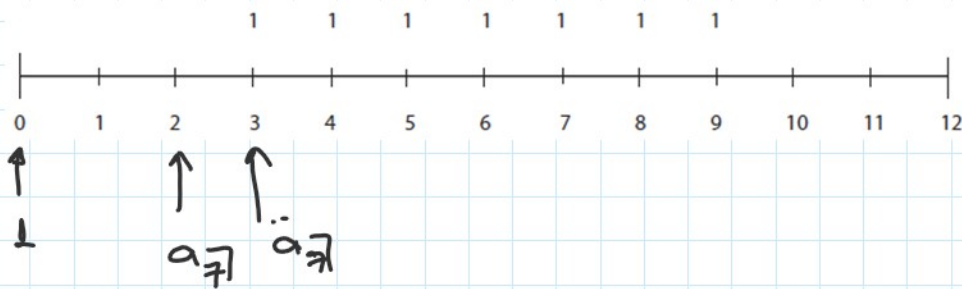
We will use a comparison date at the point where the interest rate changes. The equation of value at age 65 is

$$3000 \ddot{s}_{\overline{25}|.08} = R \ddot{a}_{\overline{15}|.07} \quad \checkmark$$

so that

$$R = \frac{3000 \ddot{s}_{\overline{25}|.08}}{\ddot{a}_{\overline{15}|.07}} = \frac{236,863.25}{9.74547} = \$24,305$$

to the nearest dollar.



Case 1: $a_{77} \cdot v^2$

$2 \mid a_{77}$

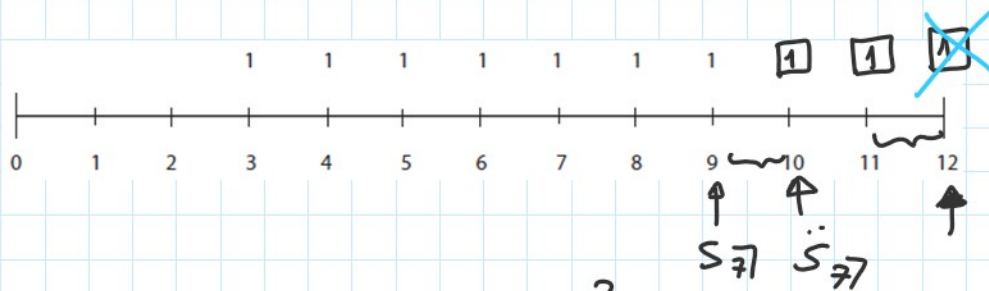
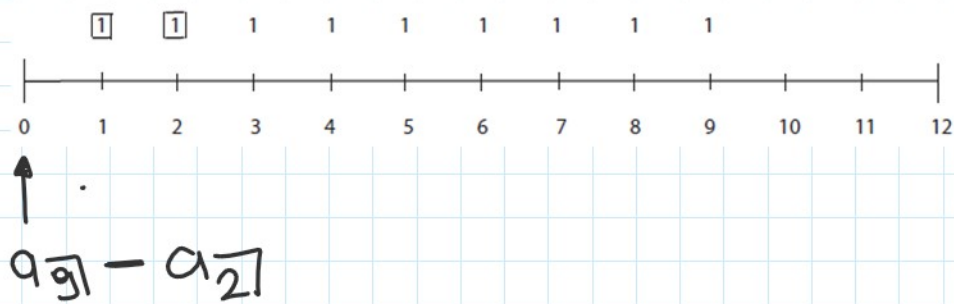
$= PV$

Ann. due:

$\ddot{a}_{77} \cdot v^3$

$3 \mid \ddot{a}_{77}$

$= PV$

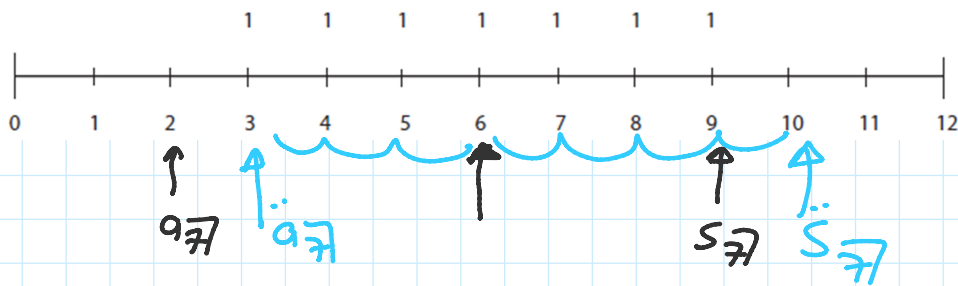


Ann. imm $\Rightarrow s_{77} \cdot (1+i)^3$

Ann. due $\Rightarrow \ddot{s}_{77} \cdot (1+i)^2$

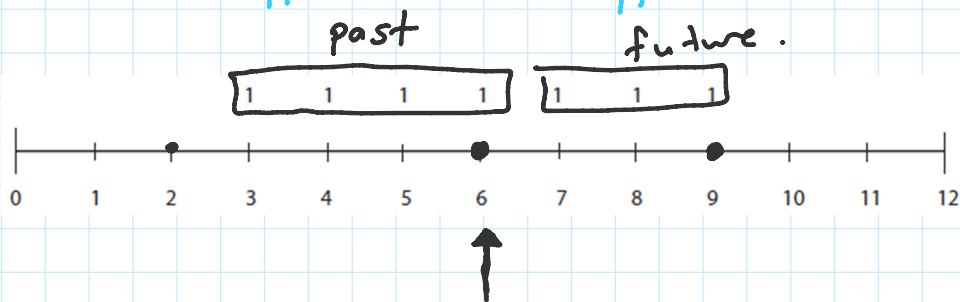
imaginary Ann imm $\Rightarrow s_{107} - s_{37}$

" " due $\Rightarrow \ddot{s}_{97} - \ddot{s}_{27}$

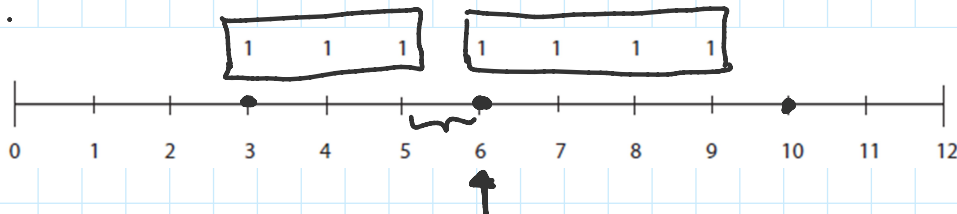


$$a_7 \cdot (1+i)^4 = s_7 \cdot v^3$$

$$\ddot{a}_7 \cdot (1+i)^3 = \ddot{s}_7 \cdot v^4$$



$$\text{Ann-imm} \Rightarrow s_7 + a_3$$



$$\text{Ann. due} \Rightarrow \ddot{s}_3 + \ddot{a}_4$$